Assignment 4

Using the Grade 9 dataset are there differences in the mean number of hours playing video games for those students who report 8 or more hours of sleep compared to those with less than 8 hours of sleep? Use Q81 and treat it as a continuous variable. Report the appropriate statistics.

A total of 2008 subjects reported getting < 8 hours of sleep and a total of 1013 subjects reported getting >= 8 hours of sleep. Because the two groups have >> 30 samples, the Central Limit Theorem applies, and we will answer the posed question by using a 2-sample T-test.

H_O: the population mean of the # of hours playing video games for those students who report 8 or more hours of sleep = the population mean of the # of hours playing video games for those students who report less than 8 hours of sleep
H_A: the population mean of the # of hours playing video games for those students who report 8 or more hours of sleep ≠ the population mean of the # of hours playing video games for those students who report less than 8 hours of sleep

Let's first carry out the folded F-test.
H_O: variance in # of hours playing video games in students who report 8 or more hours of sleep = variance in # of hours playing video games in students who less than 8 hours of sleep
H_A: variance in # of hours playing video games in students who report 8 or more hours of sleep ≠ variance in # of hours playing video games in students who less than 8 hours of sleep

The folded-F test is significant (p = 0.0472) at an alpha level of 5%, so we reject H_O and conclude that the variance in # of hours playing video games in students who report 8 or more hours of sleep ≠ variance in # of hours playing video games in students who less than 8 hours of sleep. Thus, we will use the **Satterthwaite method**.

Our T-statistic is 6.02, with a corresponding p-value of <0.0001. We therefore reject H_O and conclude that the population mean of the # of hours playing video games for those students who report 8 or more hours of sleep **is not equal to** the population mean of the # of hours playing video games for those students who report less than 8 hours of sleep.
In our data:

- Mean # of hours playing video games among those students who report 8 or more hours of sleep: 3.88 hrs (standard deviation 2.05 hrs; range 1-7 hrs)
- Mean # of hours playing video games among those students who report less than 8 hours of sleep: 4.38 hrs (standard deviation 2.16 hrs; range 1-7 hrs)
- The mean difference between the two groups (Mean # of hours playing video games among those students who report less than 8 hours of sleep - Mean # of hours playing video games among those students who report 8 or more hours of sleep) is 0.49 hrs (95% 0.33 – 0.65). Note that since the 95% CI excludes the null value of zero, we know our results are significant (at an alpha level of 5%).

Table 2. Selected characteristics of high school students grades 10-12 by short sleep duration on average school night–Youth Risk Behavior Survey 2017

Characteristic	Total N = 8691 n (%[a])	Short Sleep Duration		*p*-value
		Yes (n=6802(78.26)) n (%[a])	No (n=1889(21.74)) n (%[a])	
Video/Computer games				<0.0001[b]
<3 hrs per day	4764(57.12)	3620 (55.39)	1144(63.41)	
≥3 hrs per day	3576(42.88)	2916(44.61)	660(36.59)	
Missing	351	266	85	
Sex				0.5823[b]
Female	4476(51.62)	3513(51.78)	963(51.06)	
Male	4195(48.38)	3272(48.22)	923(48.94)	
Missing	20	17	3	
Grade				<0.0001[b]
10	2968(34.15)	2187(32.15)	781(41.34)	
11	2940(33.83)	2308(33.93)	632(33.46)	
12	2783(32.02)	2307(33.92)	476(25.20)	
Race/Ethnicity				<0.0001[b]
White	3730(43.50)	2858(42.56)	872(46.91)	
Black	1468(17.12)	1150(17.13)	318(17.11)	
Asian	403(4.70)	347(5.17)	56(3.01)	
Hispanic/Multiple Hispanic	2375(27.70)	1870(27.85)	505(27.17)	
Other	598(6.97)	490(7.30)	108(5.81)	
Missing	117	87	30	

[a]Percent based on known values;
[b]Obtained via the Chi-squated test;

Methods:

Two YRBS 2017 datasets were imported into SAS and combined into a single dataset. Duplicate observations were removed using ID as the unique identifier. Students were categorized as having short sleep duration (<8 hours) or adequate sleep (≥8 hours) based on reported hours of sleep on an average school night. The dataset was restricted to students in grades 10–12 with

non-missing sleep data. Video/computer game use was categorized as <3 hours per day versus ≥3 hours per day. Reported race/ethnicity was collapsed into the following categories: White, Black, Asian, Hispanic/multiple Hispanic, and Other. Chi-square tests were used to examine associations between short sleep duration and selected demographic and behavioral characteristics. Statistical significance was assessed at $\alpha = 0.05$.

Results:

Among students in grades 10–12 (N=8,691), 78.3% reported short sleep duration (<8 hours) and 21.7% reported sleeping ≥8 hours on an average school night.

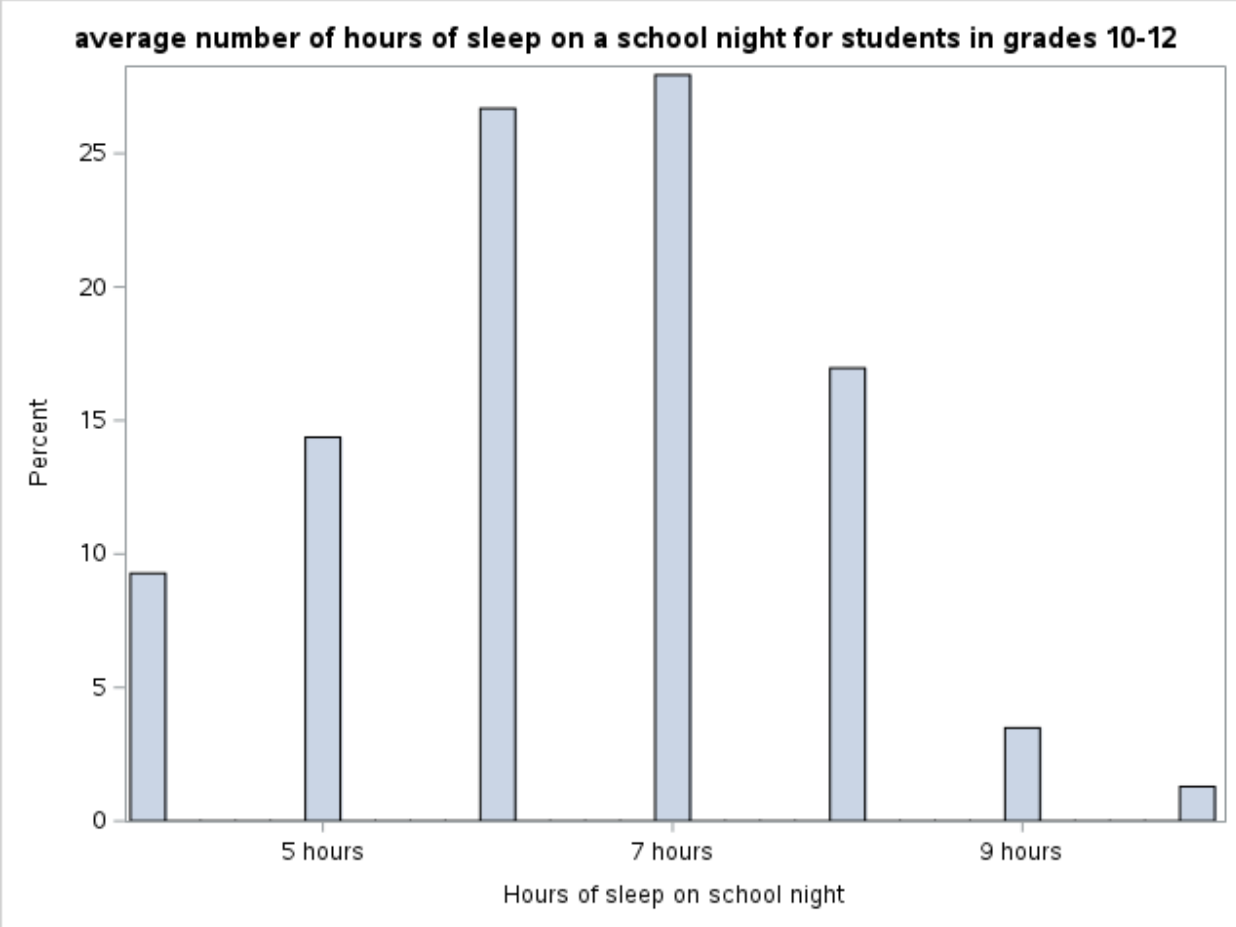
Short sleep was significantly associated with video/computer game use ($\chi^2=37.21$, $p<0.0001$). Of those in the short sleep group, 44.6% reported spending ≥3 hrs per day playing video/computer games; of those who slept ≥8 hours/night, 36.6% reported spending ≥3 hrs per day playing video/computer games.

Sex was not significantly associated with short sleep duration ($\chi^2=0.3025$, $p=0.5823$).

Grade level was found to be significantly associated with sleep duration ($\chi^2=71.77$, $p<0.0001$). Of those sleeping ≥8 hours/night, the majority were in the 10th grade (41.34%). The proportion of short sleepers was approximately equal between the 10th, 11th, and 12th grades (32%, 34%, and 34% respectively).

Race/ethnicity was also significantly associated with short sleep ($\chi^2=25.57$, $p<0.0001$). Of the short sleepers, the majority were White (42.56%), followed by Hispanic/Multiple Hispanic (27.7%), and Black students (17.12%). A similar pattern was observed among those who reported to be sleeping ≥8 hours – the majority were White (46.91%), followed by Hispanic/Multiple Hispanic (27.17), and Black (17.11%).

Histogram



Log for Assignment 4:

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69         *Assignment 4;
70
71         proc import
datafile="/home/u64319633/SAS2/Week5/yrbswave1_ID.sav"
72         DBMS=sav out=yrbs1 replace;
73
NOTE: One or more variables were converted because the data type
is not supported by the V9 engine. For more details, run with
      options MSGLEVEL=I.
NOTE: The import data set has 6500 observations and 242
variables.
NOTE: WORK.YRBS1 data set was successfully created.
```

```
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.06 seconds
      user cpu time       0.04 seconds
      system cpu time     0.02 seconds
      memory              4367.71k
      OS Memory           32376.00k
      Timestamp           03/02/2026 07:53:20 PM
      Step Count                        261  Switch Count  4
      Page Faults                       0
      Page Reclaims                     529
      Page Swaps                        0
      Voluntary Context Switches        30
      Involuntary Context Switches      5
      Block Input Operations            0
      Block Output Operations           25592

74         proc import
datafile="/home/u64319633/SAS2/Week5/yrbswave1B_ID.sav"
75         dbms=sav out=yrbs1b replace;
76
NOTE: One or more variables were converted because the data type
is not supported by the V9 engine. For more details, run with
      options MSGLEVEL=I.
NOTE: The import data set has 8320 observations and 242
variables.
NOTE: WORK.YRBS1B data set was successfully created.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.07 seconds
      user cpu time       0.06 seconds
      system cpu time     0.01 seconds
      memory              4255.75k
      OS Memory           32376.00k
      Timestamp           03/02/2026 07:53:20 PM
      Step Count                        262  Switch Count  4
      Page Faults                       0
      Page Reclaims                     529
      Page Swaps                        0
      Voluntary Context Switches        28
      Involuntary Context Switches      6
      Block Input Operations            0
      Block Output Operations           32520

77         proc contents data=yrbs1;
78         run;
NOTE: PROCEDURE CONTENTS used (Total process time):
      real time           0.17 seconds
      user cpu time       0.18 seconds
```

```
      system cpu time      0.00 seconds
      memory               3521.59k
      OS Memory            31160.00k
      Timestamp            03/02/2026 07:53:20 PM
      Step Count                          263  Switch Count  0
      Page Faults                         0
      Page Reclaims                       283
      Page Swaps                          0
      Voluntary Context Switches          5
      Involuntary Context Switches        2
      Block Input Operations              0
      Block Output Operations             128

79          proc contents data=yrbs1b;
80          run;
NOTE: PROCEDURE CONTENTS used (Total process time):
      real time            0.18 seconds
      user cpu time        0.18 seconds
      system cpu time      0.00 seconds
      memory               2488.81k
      OS Memory            31160.00k
      Timestamp            03/02/2026 07:53:21 PM
      Step Count                          264  Switch Count  0
      Page Faults                         0
      Page Reclaims                       285
      Page Swaps                          0
      Voluntary Context Switches          3
      Involuntary Context Switches        2
      Block Input Operations              0
      Block Output Operations             136

81          *1. data was read correctly;
82
83          data alldata;
84          set yrbs1 yrbs1b;
85          run;
NOTE: There were 6500 observations read from the data set
WORK.YRBS1.
NOTE: There were 8320 observations read from the data set
WORK.YRBS1B.
NOTE: The data set WORK.ALLDATA has 14820 observations and 242
variables.
NOTE: DATA statement used (Total process time):
      real time            0.04 seconds
      user cpu time        0.02 seconds
      system cpu time      0.02 seconds
      memory               5887.90k
```

```
      OS Memory           35004.00k
      Timestamp           03/02/2026 07:53:21 PM
      Step Count                        265  Switch Count  6
      Page Faults                       0
      Page Reclaims                     895
      Page Swaps                        0
      Voluntary Context Switches        27
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           57608

86
87         PROC SORT DATA= alldata
88         DUPOUT=alldata_dup NODUPKEY;
89         BY id;
90         RUN;
NOTE: There were 14820 observations read from the data set
WORK.ALLDATA.
NOTE: 55 observations with duplicate key values were deleted.
NOTE: The data set WORK.ALLDATA_DUP has 55 observations and 242
variables.
NOTE: The data set WORK.ALLDATA has 14765 observations and 242
variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.04 seconds
      user cpu time       0.00 seconds
      system cpu time     0.03 seconds
      memory              32975.03k
      OS Memory           61240.00k
      Timestamp           03/02/2026 07:53:21 PM
      Step Count                        266  Switch Count  8
      Page Faults                       0
      Page Reclaims                     7662
      Page Swaps                        0
      Voluntary Context Switches        41
      Involuntary Context Switches      1
      Block Input Operations            0
      Block Output Operations           58128

91         PROC print data = alldata_dup; run;
NOTE: There were 55 observations read from the data set
WORK.ALLDATA_DUP.
NOTE: PROCEDURE PRINT used (Total process time):
      real time           1.09 seconds
      user cpu time       1.08 seconds
      system cpu time     0.01 seconds
      memory              2708.68k
```

```
      OS Memory           30132.00k
      Timestamp           03/02/2026 07:53:22 PM
      Step Count                        267  Switch Count  0
      Page Faults                       0
      Page Reclaims                     61
      Page Swaps                        0
      Voluntary Context Switches        0
      Involuntary Context Switches      20
      Block Input Operations            0
      Block Output Operations           800

92
93          DATA grade9  grades10_12;
94          SET alldata;
95          IF q3=1 THEN OUTPUT grade9;
96          ELSE IF q3 in (2 3 4) THEN OUTPUT grades10_12;
97          Run;
NOTE: There were 14765 observations read from the data set
WORK.ALLDATA.
NOTE: The data set WORK.GRADE9 has 3921 observations and 242
variables.
NOTE: The data set WORK.GRADES10_12 has 10700 observations and
242 variables.
NOTE: DATA statement used (Total process time):
      real time           0.03 seconds
      user cpu time       0.01 seconds
      system cpu time     0.02 seconds
      memory              5879.00k
      OS Memory           35516.00k
      Timestamp           03/02/2026 07:53:22 PM
      Step Count                        268  Switch Count  8
      Page Faults                       0
      Page Reclaims                     877
      Page Swaps                        0
      Voluntary Context Switches        43
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           57104

98
99          DATA grade9_new;
100         set grade9;
101         IF Q88 IN ( 1 2 3 4 ) THEN SHORTSLEEP = 0;
102           ELSE IF Q88 IN ( 5 6 7 ) THEN SHORTSLEEP = 1;
103         IF Q88 NOT EQ .;
104           RUN;
```

```
NOTE: There were 3921 observations read from the data set
WORK.GRADE9.
NOTE: The data set WORK.GRADE9_NEW has 3021 observations and 243
variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      user cpu time       0.01 seconds
      system cpu time     0.01 seconds
      memory              4234.09k
      OS Memory           32952.00k
      Timestamp           03/02/2026 07:53:22 PM
      Step Count                        269  Switch Count  3
      Page Faults                       0
      Page Reclaims                     528
      Page Swaps                        0
      Voluntary Context Switches        16
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           12040

105
106        proc ttest data=grade9_new;
107        class shortsleep;
108        var q81;
109        run;
NOTE: PROCEDURE TTEST used (Total process time):
      real time           0.39 seconds
      user cpu time       0.20 seconds
      system cpu time     0.07 seconds
      memory              18772.34k
      OS Memory           41568.00k
      Timestamp           03/02/2026 07:53:22 PM
      Step Count                        270  Switch Count  59
      Page Faults                       0
      Page Reclaims                     28670
      Page Swaps                        0
      Voluntary Context Switches        1116
      Involuntary Context Switches      10
      Block Input Operations            0
      Block Output Operations           5408

110
111        data grades10_12new;
112        set grades10_12;
113        IF Q88 IN ( 1 2 3 4 ) THEN SHORTSLEEP = 0;
114          ELSE IF Q88 IN ( 5 6 7 ) THEN SHORTSLEEP = 1;
115        IF Q88 NOT EQ .;
```

```
116         If q81=. then videogame=.;
117         else if q81 in (1 2 3 4) then videogame=1;
118         else if q81 in (5 6 7) then videogame=2;
119         if raceeth=. then race=.;
120         else if raceeth=5 then race=1;
121         else if raceeth=3 then race=2;
122         else if raceeth=2 then race=3;
123         else if raceeth in (6 7) then race=4;
124         else if raceeth in (1 4 8) then race=5;
125         RUN;
NOTE: There were 10700 observations read from the data set
WORK.GRADES10_12.
NOTE: The data set WORK.GRADES10_12NEW has 8691 observations and
245 variables.
NOTE: DATA statement used (Total process time):
      real time           0.03 seconds
      user cpu time       0.02 seconds
      system cpu time     0.02 seconds
      memory              4371.81k
      OS Memory           37564.00k
      Timestamp           03/02/2026 07:53:22 PM
      Step Count                        271  Switch Count  5
      Page Faults                       0
      Page Reclaims                     530
      Page Swaps                        0
      Voluntary Context Switches        19
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           33800

126
127         Proc format;
128
128       !  value sleepfmt
129         0="less than 8 hours"
130         1="8 or more";
NOTE: Format SLEEPFMT is already on the library WORK.FORMATS.
NOTE: Format SLEEPFMT has been output.
131             value gamefmt
132          1= "less than 3"
133          2= "3 or more";
NOTE: Format GAMEFMT is already on the library WORK.FORMATS.
NOTE: Format GAMEFMT has been output.
134
134       !  value sexfmt
135         1="female"
136         2="male";
```

```
NOTE: Format SEXFMT is already on the library WORK.FORMATS.
NOTE: Format SEXFMT has been output.
136      !            *var q2;
137
137      !  value gradefmt
138        2="10th grade"
139        3="11th grade"
140        4="12th grade";
NOTE: Format GRADEFMT is already on the library WORK.FORMATS.
NOTE: Format GRADEFMT has been output.
140      !                   *var q3;
141
141      !  value racefmt
142        1="white"
143        2="black"
144        3="asian"
145        4="hispanic or multiple hispanic"
146        5="other";
NOTE: Format RACEFMT is already on the library WORK.FORMATS.
NOTE: Format RACEFMT has been output.
147        run;
NOTE: PROCEDURE FORMAT used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              472.18k
      OS Memory           34484.00k
      Timestamp           03/02/2026 07:53:22 PM
      Step Count                        272  Switch Count  0
      Page Faults                       0
      Page Reclaims                     15
      Page Swaps                        0
      Voluntary Context Switches        0
      Involuntary Context Switches      1
      Block Input Operations            0
      Block Output Operations           24

148        proc freq data=grades10_12new;
149        tables shortsleep;
150        format shortsleep sleepfmt.;
151        run;
NOTE: There were 8691 observations read from the data set
WORK.GRADES10_12NEW.
NOTE: PROCEDURE FREQ used (Total process time):
      real time           0.01 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
```

```
      memory              2399.56k
      OS Memory           35772.00k
      Timestamp           03/02/2026 07:53:22 PM
      Step Count                        273  Switch Count  3
      Page Faults                       0
      Page Reclaims                     311
      Page Swaps                        0
      Voluntary Context Switches        17
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264

152         Proc freq data=grades10_12new;
153         tables videogame*shortsleep/All expected;
154         tables q2*shortsleep/All expected;
155         tables q3*shortsleep/all expected;
156         tables race*shortsleep/all expected;
157             format shortsleep sleepfmt. videogame gamefmt. q2
sexfmt. q3 gradefmt. race racefmt.;
158         run;
NOTE: There were 8691 observations read from the data set
WORK.GRADES10_12NEW.
NOTE: PROCEDURE FREQ used (Total process time):
      real time           0.18 seconds
      user cpu time       0.18 seconds
      system cpu time     0.01 seconds
      memory              3050.15k
      OS Memory           36288.00k
      Timestamp           03/02/2026 07:53:22 PM
      Step Count                        274  Switch Count  6
      Page Faults                       0
      Page Reclaims                     443
      Page Swaps                        0
      Voluntary Context Switches        32
      Involuntary Context Switches      5
      Block Input Operations            0
      Block Output Operations           608

159
160         title "average number of hours of sleep on a school
night for students in grades 10-12";
161
161       !  proc sgplot data=grades10_12new;
162         histogram q88;
163         run;
NOTE: PROCEDURE SGPLOT used (Total process time):
      real time           0.07 seconds
```

```
      user cpu time       0.03 seconds
      system cpu time     0.00 seconds
      memory              3132.65k
      OS Memory           36672.00k
      Timestamp           03/02/2026 07:53:23 PM
      Step Count                        275  Switch Count  7
      Page Faults                       0
      Page Reclaims                     602
      Page Swaps                        0
      Voluntary Context Switches        165
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           1024

NOTE: There were 8691 observations read from the data set
WORK.GRADES10_12NEW.
164        *4) There is a mean difference of 0.4937(95% CI:
0.33, 0.65) hours playing video games for those students who
165        report 8 or more hours of sleep compared to those
with less than 8 hours of sleep. With the confidence interval
166        (95% CI: 0.33, 0.65) which excludes 0 and the pvalue
of <0.0001, we can conclude that there is a significant
167        difference of mean video game hours played between
9th graders that sleep 8 or more hours versus those who sleep
less
167      ! than 8 hours.
168
```